

Course: Information Assurance and Security

Lab Title: Assignment#1a

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Date:

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Case Study 1: BTK Killer (Dennis Rader)

(From: www.britannica.com/biography/Dennis-Rader)

Case Type: Criminal (Serial Murder)

Environment: Floppy disk, Windows 98 PC

Industry: Law Enforcement

Method/Appliance Involved:

- **FTK Imager** (disk imaging)
- **EnCase** (metadata analysis)
- **Hex editor** (file carving)

Method of Discovery:

- Rader's habit of leaving cryptic clues led investigators to search his computer, which ultimately exposed him.
- Killer mailed floppy disk to police, believing metadata would be erased

Method of Solving the Problem:

- Imaged the disk (MD5 hash: d41d8cd98f00b204e9800998ecf8427, verified).

Recovered critical files:

- **Christ Lutheran Church.rtf** (contained "Dennis" in metadata)
- **BAD.rtf** (deleted confession draft)

Analyzed metadata:

File created: 2004-02-16 | **Last modified:** 1991-03-17 (provenance anomaly)

Linked evidence:

- Metadata matched Rader's church computer
- Paper documents matched disk files

Outcome: Rader's confession confirmed the evidence on the disk he sent, ultimately securing his conviction for multiple murders.

Case Study 2: Waymo vs. Uber

(From: <https://www.courtlistener.com/docket/4609586/waymo-llc-v-uber-technologies-inc/>)

Case Type: IP Theft (Extortion Templates)

Environment: Corporate laptops, GitHub repos

Industry: Cybercrime

Method/Appliance Involved:

- Cellebrite (device imaging)
- GitGuardian (code plagiarism detection)

Method of Discovery:

Ex-employee downloaded 14,000 files pre-Uber

Method of Solving the Problem:

- Imaged employee's laptop (verified SHA-1 hash).

Recovered critical files:

- `lidar_design.zip` (last accessed pre-employment)
- `waymo_passwords.txt` (in Recycle Bin)

Analyzed metadata:

Git commits showed stolen code 2 days post-hire

Linked evidence:

Code matched Waymo's proprietary designs

Outcome: Uber settled with Waymo in 2018, agreeing to pay approximately **\$245 million in equity** and to **not use Waymo's trade secrets**, ending the trial just four days in. The case highlighted the importance of safeguarding intellectual property in the autonomous vehicle industry.

Case Study 3: Silk Road

(From: DAJAO & CUIZON -- Digital Forensic Investigation Report --Drug Case.pdf)

Case Type: Drug/Narcotics Trafficking

Environment: Tor, Bitcoin blockchain

Industry: Cybercrime

Method/Appliance Involved:

- **Chainalysis** (Bitcoin tracing)
- **Wireshark** (IP leak detection)

Method of Discovery:

- Ulbricht used his personal email (rossulbricht@gmail.com) on Stack Overflow while asking for coding help—an early digital trail that helped link him to the Silk Road.

Method of Solving the Problem:

1. **Imaged Silk Road servers** (verified the hashes).
2. **Recovered critical data:**
 - `wallet.dat` (Bitcoin private keys)
 - `users.sqlite` (usernames/IP logs)
3. **Analyzed metadata:**
 - Bitcoin tx `a1f4...` led to Ulbricht's wallet
4. **Linked evidence:**
 - IP logs matched his café login sessions

Outcome: Digital evidence linked Ulbricht to the Silk Road marketplace, confirming his role in facilitating large-scale drug trafficking operations.

Case Study 4: Bangladesh Bank Heist

(From:

<https://www.isaca.org/resources/isaca-journal/issues/2023/volume-6/lessons-learned-from-the-bangladesh-bank-heist?>)

Case Type: Financial Fraud

Environment: SWIFT network, Windows servers

Industry: Cybersecurity, Banking

Method/Appliance Involved:

- Volatility (malware RAM analysis)
- Splunk (log correlation)

Method of Discovery:

- Typo in \$81M transfer request (misspelled "foundation")

Method of Solving the Problem:

Imaged infected servers (Verified the hashes).

1. Recovered critical files:

- `swift_gateway.dll` (malware binary)
- `print_job.ini` (C&C config)

2. Analyzed metadata:

- **Malware compiled 2015-12-25** (matched Lazarus Group patterns)

3. Linked evidence:

- IPs traced to North Korean hacker apartments

Outcome: Hackers successfully transferred \$81 million via SWIFT, exposing critical vulnerabilities in international banking security and prompting global reforms in cyber risk management.

Case Study 5: Enron Scandal

(From: <https://www.britannica.com/event/Enron-scandal?>)

Case Type: Business Crime (Evidence Concealment)

Environment: Exchange servers, PST files

Industry: Energy

Method/Appliance Involved:

- Forensic Email Examiner (PST analysis)
- Nuix (keyword search)

Method of Discovery:

- Whistleblower reported "off-book" partnerships

Method of Solving the Problem:

1. **Imaged 120 employee workstations** (hashes were verified).
2. **Recovered critical emails:**
 - `Raptor_Deal.doc` (hidden losses)
 - `Skilling_Resign.txt` (deleted draft)
3. **Analyzed metadata:**
 - 500+ emails with "delete after reading" flags
4. **Linked evidence:**
 - Handwritten notes matched email content

Outcome: Enron declared bankruptcy in 2001, wiping out over \$74 billion in shareholder value, leading to criminal convictions of top executives, the collapse of Arthur Andersen (one of the "Big Five" accounting firms), and the creation of the Sarbanes-Oxley Act of 2002 to improve corporate accountability and financial transparency.